

BILAL AHMED

Robotics Software & Autonomy Engineer | Field Robotics & Autonomous Systems

PROFILE

Erasmus Mundus MSc researcher in Intelligent Field Robotic Systems with hands-on experience developing autonomous systems across underwater, mobile, legged and manipulation platforms. Strong in ROS/ROS 2, C++/Python, localization and SLAM, navigation, multimodal perception and robot learning, with real-robot AUV deployment, peer-reviewed research and quantitatively evaluated autonomy systems.

EDUCATION

Erasmus Mundus Joint Master in Intelligent Field Robotic Systems 2024 - Present

University of Girona | University of Zagreb | Heriot-Watt University

GPA: 9.40/10.0 | Focus: autonomous systems, perception, localization, planning and robot learning.

Bachelor of Mechatronics Engineering 2018 - 2022

Mehran University of Engineering and Technology, Jamshoro

CGPA: 3.9/4.0 | **2nd in Faculty** | Mechatronics, control, embedded systems and robotics.

TECHNICAL SKILLS

Robot Software: ROS/ROS 2, C++, Python, Nav2, Behavior Trees, FastAPI, MQTT, testing and debugging

Localization & Planning: EKF, particle filters, visual odometry, SLAM, GTSAM pose graphs, GICP, path planning, exploration

Perception: OpenCV, stereo vision, sonar, LiDAR, point clouds, PCL, Open3D, FoundationStereo, Depth Anything V2, DINOv2

Robot Learning & AI: PyTorch, PPO, Stable-Baselines3, domain randomization, behavior cloning, VLA/VLM/LLM systems, RAG, FAISS

Simulation & Evaluation: MuJoCo, Gazebo, StoneFish, Panda3D, Gymnasium, seeded benchmarks, statistical evaluation

RESEARCH EXPERIENCE

Visiting Scholar - Agentic Recovery for Marine Robotics 2026 - Present

Ocean Systems Lab, Heriot-Watt University

- Architected MAESTRO, a multi-agent ROS 2 framework combining residual-based anomaly evidence, fault diagnosis, mission-impact reasoning, RAG-supported recovery generation and operator approval for autonomous underwater missions.
- Evaluated 255 fault-recovery scenarios: achieved 89% recovery-decision accuracy, improved Coordinated Success Score from 2.42 to 3.92, and obtained 80% validated success across 50 generated ROS 2 nodes.

Research Intern - Multimodal Underwater 3D Reconstruction 2025

COE MARBLE Lab, University of Zagreb

- Developed a stereo-sonar-AUV-pose fusion pipeline using FoundationStereo, sonar feature extraction, particle-filter height estimation and point-cloud fusion for offshore-structure reconstruction.
- Investigated probabilistic reduction of forward-looking sonar vertical ambiguity and presented the work at Breaking the Surface 2025.

Volunteer Researcher - MiniGirona Autonomous Underwater Vehicle 2024 - 2025

CIRS Lab, University of Girona

- Contributed to mechanical-sonar initialization, DVL/INS/sonar EKF localization, behavior-tree mission execution, stereo perception and task-priority manipulation on the real MiniGirona I-AUV.
- Integrated and validated autonomy modules in simulation and real-robot missions; team placed 2nd in the RAMI 2025 Underwater Inspection & Maintenance Challenge, with related work published at ROBOT 2025.

SELECTED ENGINEERING & RESEARCH PROJECTS

TIAGo ROS 2 Navigation Integration

2026

Mobile Robotics | ROS 2, Nav2, C++, Python, FastAPI, MQTT, Gazebo

- Built a C++ Nav2 integration adapter with ROS, REST and MQTT mission interfaces, readiness gating, pause/resume/cancel control, health monitoring, stuck detection and structured commissioning evidence.
- Completed 3/3 commissioning missions with 8.53 s mean navigation time and 0.23 m mean final position error; explicitly reported the remaining 0.10 m acceptance-target tuning gap.

Adaptive Go2 Locomotion under Dynamics Shift

2026

Legged Robotics | MuJoCo, PPO, PyTorch, Domain Randomization

- Designed nominal, domain-randomized and context-conditioned PPO controllers for Unitree Go2 velocity tracking under mass, friction, payload, actuator, latency, terrain and disturbance shifts.
- Evaluated 3 frozen policies across 47 scenarios and 1,410 seeded episodes; DR-PPO reduced mean OOD tracking RMSE by 22.3%, with 86.5% aggregate benchmark success and 25 automated tests.

Failure-Aware Multimodal Franka Manipulation

2026

Manipulation | PyTorch, MuJoCo, Behavior Cloning, Force/Torque Sensing

- Built a 200-demonstration multimodal behavior-cloning pipeline with RGB, robot state and force/torque inputs, plus online failure detection and retry-limited recovery skills for contact-rich peg insertion.
- Detected a controlled object slip, autonomously regrasped and completed insertion while remaining below a 60 N safety limit; validated the implementation with Ruff, mypy and 16 automated tests.

BathyGraph-Lite - Bathymetric Pose-Graph SLAM

2026

Marine Robotics | GTSAM, Open3D, Generalized ICP, AUVLib

- Built a GPS-denied AUV SLAM pipeline from 97,318 dead-reckoning states and 36.1M multibeam points, associating 296 measured submaps through DR-proximity candidates, GICP registration and Huber-robust pose-graph optimization.
- Improved accepted-edge bathymetric overlap consistency by 6.3% and reduced accepted-pair vertical RMS consistency from 2.219 m to 1.466 m after graph optimization.

Underwater Metric Depth Estimation

2025 - 2026

Perception | PyTorch, Depth Anything V2, Monodepth2, FLSea

- Built a leakage-free metric-depth benchmark with strict scene-level separation, comparing supervised, self-supervised and stereo approaches under canyon-to-canyon underwater domain shift.
- Reduced held-out AbsRel from 0.1995 to 0.1509 (24.35%), reduced RMSE from 0.8623 m to 0.7593 m, and improved delta1 from 0.7148 to 0.8111.

PUBLICATIONS & RESEARCH COMMUNICATION

- **Peer-reviewed paper:** "Advancing Accessible Underwater Robotics: The Mini-Girona I-AUV at RAMI 2025" - ROBOT 2025, Porto, Portugal.
- **Master thesis:** "MAESTRO: Agentic Fault Diagnosis and Recovery for Autonomous Underwater Robots" - Heriot-Watt University, 2026.
- **Research presentation:** "Multimodal 3D Reconstruction of Offshore Structures" - Breaking the Surface 2025, Cyprus.

SELECTED AWARDS

- 1st Place - Research Poster Competition, RAMI Marine Robots Competition, La Spezia, Italy (2025)
- 2nd Place - Underwater Inspection & Maintenance Challenge, RAMI Marine Robots Competition, La Spezia, Italy (2025)
- 2nd Prize - AI Agent Olympics, Milan AI Week, Milan, Italy (2026)

ADDITIONAL ROBOTICS WORK

Cross-platform portfolio: AquaNav-FM underwater visual localization and waypoint prediction; AquaAdapt self-supervised underwater place recognition; MarsSim ROS 2 planetary rover simulation; humanoid PPO navigation; Unitree Go1 frontier exploration; stereo visual odometry & SLAM; multi-robot aerial coordination; OpenVLA KUKA manipulation; TIAGo multimodal HRI.